



BUILDING INVESTIGATION TIMELINES

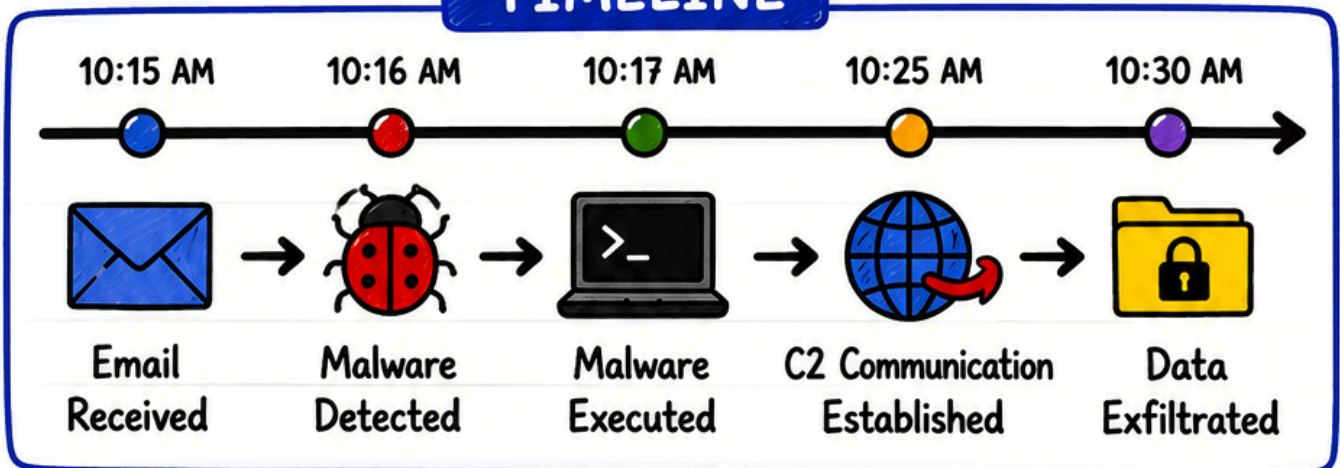


FOR SOC ANALYSTS

Turn scattered events into a clear, chronological story.

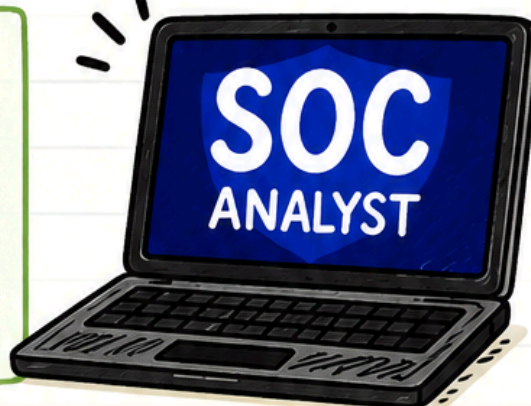


TIMELINE



BUILD TIMELINES TO:

- ✓ Understand attack sequence
- ✓ Identify key events
- ✓ Determine scope & impact
- ✓ Support incident response
- ✓ Create clear incident reports



GOOD TIMELINES = BETTER INVESTIGATIONS = STRONGER DEFENSE

WHAT IS AN INVESTIGATION TIMELINE?



An investigation timeline is a chronological list of events that occurred **before**, **during**, and **after** a security incident.



IN SIMPLE WORDS:



It turns scattered events and alerts into a clear, complete story!



A timeline helps analysts put all the puzzle pieces in order to understand what happened and when.

WHY DO SOC ANALYSTS BUILD TIMELINES?

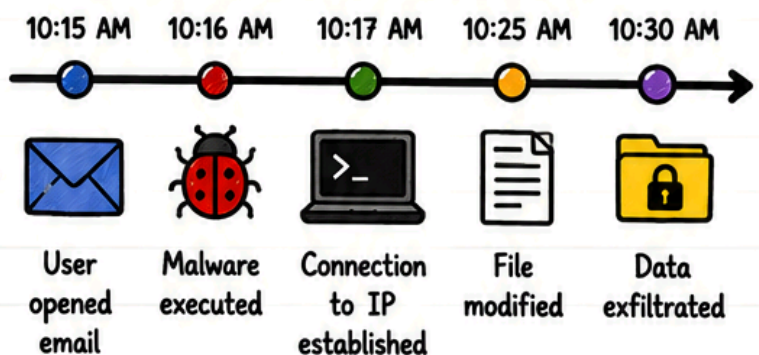
- ✓ It helps understand the full attack sequence.
- ✓ It shows how the attacker moved.
- ✓ It helps identify scope, impact, and affected systems.
- ✓ It supports accurate reporting and incident response.
- ✓ It helps prevent similar attacks in the future.



TIMELINES TURN THESE EVENTS...



...INTO A CLEAR STORY!



A timeline helps analysts see the BIG PICTURE, make better decisions, and stop threats faster!



WHY TIMELINES **MATTER**



A good timeline turns noisy alerts into a **clear, complete attack story.**

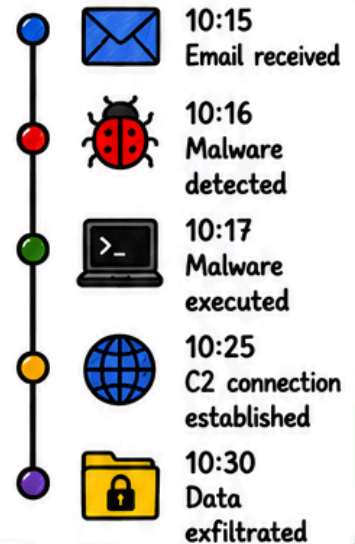
WITHOUT A TIMELINE

- ✗ Events look random and disconnected.
- ✗ Harder to understand the attacker's steps.
- ✗ Important details may be missed.
- ✗ Slower investigations and response.



WITH A TIMELINE

- ✓ Events are in chronological order.
- ✓ The attack path is easy to see.
- ✓ Helps identify root cause.
- ✓ Faster decisions and accurate reporting.



TIMELINES HELP YOU:



Connect the dots between events.



Understand the attacker's path.



Identify key actions and objectives.



Support scope, impact, and reporting.



Improve incident response and defense.

EXAMPLE: SAME ALERT, DIFFERENT VIEW

ALERT VIEW (CHAOS)

Malware executed Login from new IP

File modified Suspicious email

PowerShell run Multiple connections

Registry change Data exfiltration

TIMELINE VIEW (CLARITY)

10:15 Suspicious email received

10:16 User clicked link / Malware downloaded

10:17 Malware executed (PowerShell)

10:18 Persistence established (Registry change)

10:25 Connection to C2 server

10:30 Data exfiltration detected



Timelines bring **ORDER** to chaos and give you the **FULL PICTURE.**



THE 5-STEP TIMELINE BUILDING PROCESS



Building a strong investigation timeline follows **5 simple** but powerful steps.



**IDENTIFY
THE ANCHOR
EVENT**

**LOOK
BACKWARDS**

**LOOK
FORWARD**

**NOTE KEY
TIMESTAMPS**

**BUILD THE
NARRATIVE**



Find the event that triggered the alert.
This is your starting point.



Investigate what happened BEFORE the anchor event.
How did the attacker get in?



Investigate what happened AFTER the compromise.
Where did the attacker go?



Record important events with accurate time, system, user, and details.



Piece everything together into a clear story of what happened, step by step.

THE BIG PICTURE



Alert
Triggers



Investigate
Step by Step



Connect
All the Dots



Understand
& Report



Respond
& Defend



Follow these **5 steps** every time to turn scattered alerts into a clear attack story!

① IDENTIFY THE ANCHOR EVENT



The anchor event is the starting point of your investigation.
It is the event that triggered the alert.

WHAT TO IDENTIFY:

- ✓ What triggered the alert?
- ✓ Which system or user was affected?
- ✓ What type of activity was detected?
- ✓ Initial severity or impact

KEY POINTS:

- The anchor event becomes the reference point for your timeline.
- All events before and after this point will be investigated.
- Choose the most reliable and relevant alert.

EXAMPLES OF ANCHOR EVENTS:



Malware detected

Antivirus or EDR detected malware on a system.



Suspicious PowerShell

Unusual or malicious PowerShell activity detected.



Phishing email received

User received a suspicious email or clicked a malicious link.



Multiple failed logins

Several failed login attempts detected for a user account.



Suspicious network connection

Unusual outbound connection to a suspicious IP or domain.

ANCHOR EVENT EXAMPLE



Alert: Malware detected on workstation PC-05

Time (UTC): 10:17 AM

System: Marketing-PC-05

User: Evil.User

Severity: High

This is the starting point of our investigation!



If the starting point is correct, the entire timeline will be accurate!



② LOOK BACKWARDS

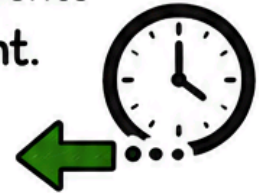
INVESTIGATE BEFORE THE ATTACK



Looking backwards means investigating events that happened **BEFORE** the anchor event.

This helps you answer:

HOW DID THE ATTACKER GET IN?



WHAT TO LOOK FOR:

- ✓ Email logs (phishing emails)
- ✓ User login activity
- ✓ Web browsing history
- ✓ File downloads
- ✓ Initial access attempts
- ✓ Privilege escalation
- ✓ Any suspicious behavior before the attack



The goal is to find the **initial access** point.

COMMON SOURCES TO CHECK:



Email Gateway / Logs

Check for malicious or suspicious emails received.



Authentication Logs

Check for successful, failed, or unusual login attempts.



Web Proxy / DNS Logs

Check for suspicious domains, downloads, or visits.



File Logs

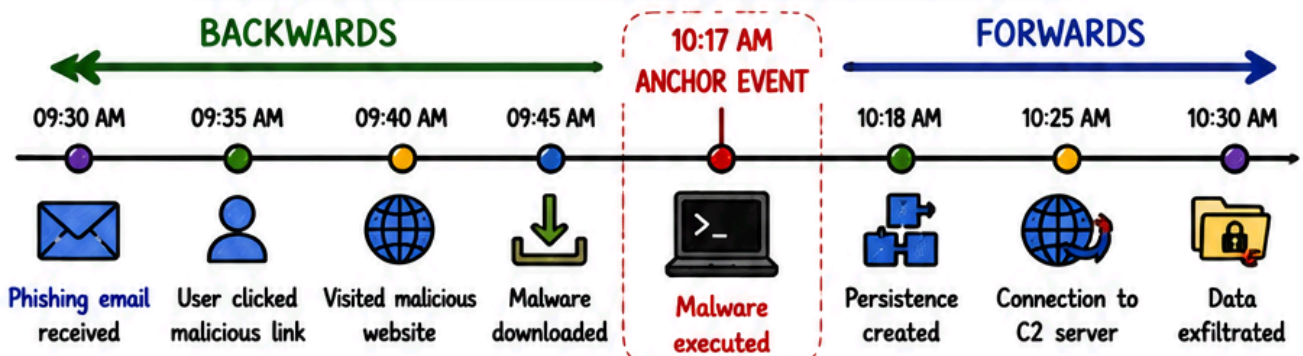
Check for downloaded or executed files before the attack.



Initial Access Techniques

Look for phishing, valid accounts, exploits, or misconfigurations.

EXAMPLE: LOOKING BACKWARDS TIMELINE



KEY TAKEAWAY:

Events before the anchor event reveal the attacker's **entry point** and early actions.

ANALYST TIP



- Go as far back as possible.
- Don't stop at the first suspicious event.
- Even small events can reveal the **attack path**!



The more you look backwards, the clearer the attacker's **entry path** becomes!



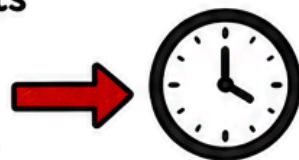
③ LOOK FORWARD INVESTIGATE AFTER THE COMPROMISE



Looking forward means investigating events that happened **AFTER** the anchor event.

This helps you answer:

WHAT DID THE ATTACKER DO NEXT?



WHAT TO LOOK FOR:

- ✓ Malware execution (process creation, scripts)
- ✓ Command and Control (C2) communications
- ✓ Persistence mechanisms created by attacker
- ✓ Lateral movement to other systems
- ✓ Data access and discovery
- ✓ Data exfiltration attempts
- ✓ Cleanup or defense evasion activities



Goal: Understand the attacker's **ACTIONS, MOVEMENT, and IMPACT** after compromise.

COMMON SOURCES TO CHECK:



Endpoint / Process Logs

Check for executed processes, scripts, or suspicious commands.



Network / Firewall Logs

Check outbound connections, C2 traffic, unusual destinations.



File Integrity / Change Logs

Check for modified, created, or new files and registries.



Active Directory / Auth Logs

Check for lateral movement, privilege escalation, new accounts.



Data Access Logs

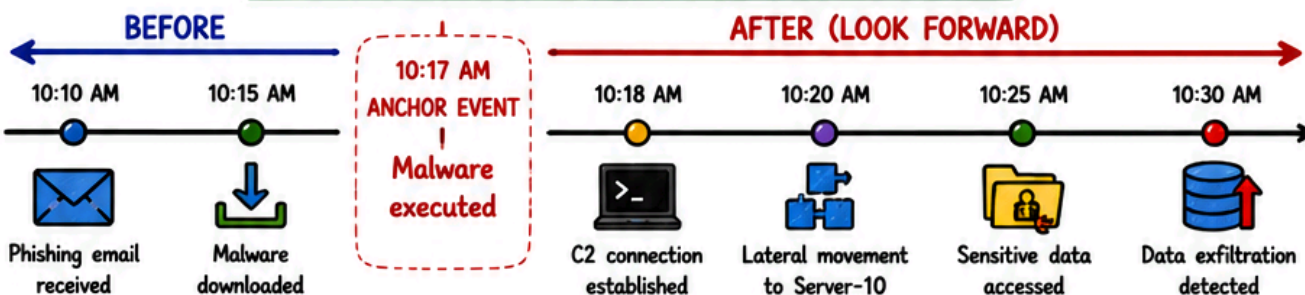
Check for large data access, sensitive file access, shares.



Security Tool Alerts

EDR, IDS/IPS, SIEM alerts after the compromise.

EXAMPLE: LOOKING FORWARD TIMELINE



ANALYST TIPS

- Follow the attacker's path **step-by-step**.
- Look for **patterns** in behavior and tools.
- Don't assume the attack stopped early.
- Keep going forward until you find the end.



The more you look forward, the clearer the impact and **attack story** becomes!



LOOK FORWARD TO FIND WHAT THE ATTACKER DID, WHERE THEY WENT, AND **WHAT THEY TOOK!**



④ NOTE KEY TIMESTAMPS



Timestamps are the backbone of every investigation timeline. They tell us WHEN each event happened.

WHAT TO RECORD FOR EACH EVENT:

TIME (UTC)



Exact date and time of the event.

EVENT



What happened?

SYSTEM



Which device or host?

USER



Who was involved?

ACTION



What action was performed?

WHY ACCURATE TIMESTAMPS MATTER:

- ✓ Helps put events in the correct order.
- ✓ Reveals the full attack path.
- ✓ Helps correlate events across systems.
- ✓ Supports accurate reporting and incident response.
- ✓ Helps prove impact, scope, and attacker dwell time.

UTC vs LOCAL TIME



UTC (Coordinated Universal Time):

- Standard time used in logs and tools.
- No confusion across time zones.
- Always use UTC for timelines!



LOCAL TIME:

- Time in your local time zone.
- Can cause confusion when correlating events.
- Convert to UTC for accuracy.

EXAMPLE TIMELINE TABLE

TIME (UTC)	EVENT	SYSTEM	USER	ACTION
10:15:22 AM	Email received	Mail-Server-01	Evil.User	Email delivered
10:16:05 AM	User clicked link	Marketing-PC-05	Evil.User	Link clicked
10:17:11 AM	Malware executed	Marketing-PC-05	Evil.User	Malware run (PowerShell)
10:18:33 AM	Persistence created	Marketing-PC-05	Evil.User	Registry added
10:25:44 AM	C2 connection	Marketing-PC-05	Evil.User	Outbound to C2 IP
10:30:52 AM	Data exfiltration	Marketing-PC-05	Evil.User	Sensitive data sent



KEY TAKEAWAY:

If your timestamps are wrong, your timeline will be WRONG!



GOOD TIMESTAMPS
= ACCURATE TIMELINE
= CLEAR ATTACK STORY



Always collect, verify, and use UTC timestamps for a reliable and professional investigation!



5 BUILD THE NARRATIVE



The narrative turns your timeline into a clear, step-by-step story of what happened.

WHAT MAKES A STRONG NARRATIVE?



ACCURATE

Based on verified evidence and timestamps.



COMPLETE

Covers the full attack from start to finish.



CLEAR

Easy to follow and understand by others.



FOCUSED

Stays on facts, avoids noise and irrelevant details.



INSIGHTFUL

Explains impact, attacker actions, and key outcomes.

STEPS TO BUILD THE NARRATIVE

- 1 FOLLOW THE TIMELINE**
Start from the anchor event and move forward in time.
- 2 DESCRIBE WHAT HAPPENED**
Explain each key event in simple, clear language.
- 3 ADD CONTEXT**
Include how, where, and why it happened (based on evidence).
- 4 HIGHLIGHT IMPACT**
Explain what data, systems, or users were affected.
- 5 CONNECT THE DOTS**
Show how each step led to the next and the overall outcome.

WHAT TO INCLUDE



EXECUTIVE SUMMARY

Brief overview of the incident and key findings.



ATTACKER JOURNEY

Step-by-step actions of the attacker from start to finish.



TECHNIQUES & TOOLS

TTPs, tools, and methods used by the attacker.



IMPACT

What was affected, data accessed, or exfiltrated.



DETECTION & RESPONSE

How it was detected and what response actions were taken.

EXAMPLE: NARRATIVE IN 5 STEPS

①



Phishing email received by user (10:15 AM).

②



User clicked link → Malware executed (10:17 AM).

③



Lateral movement to Server-10 (10:20 AM).

④



Sensitive data accessed and staged (10:25 AM).

⑤



Data exfiltrated to external IP (10:30 AM).



A strong narrative tells the story, proves the impact, and drives better decisions!



⑥ FINAL TIPS & COMMON PITFALLS



A well-built timeline is the foundation of a strong investigation and a successful outcome.

BEST PRACTICES

- ✓ **START EARLY**
Start your timeline as soon as you detect an incident.
- ✓ **USE RELIABLE SOURCES**
Use trusted logs and tools.
- ✓ **BE THOROUGH**
Collect both backward and forward events.
- ✓ **USE UTC TIMESTAMPS**
Always use UTC for consistency and accuracy.
- ✓ **DOCUMENT EVERYTHING**
Record your findings, sources, and assumptions.
- ✓ **REVIEW & VALIDATE**
Verify your timeline and cross-check key events.
- ✓ **KEEP IT UPDATED**
Update the timeline as new information comes in.

COMMON MISTAKES TO AVOID



WRONG TIMESTAMPS

Using local time or inconsistent timezones leads to errors.



MISSING EVENTS

Skipping small details can hide critical parts of the attack.



ONLY LOOKING FORWARD

Not checking what happened before can miss the attacker's entry point.



NO CONTEXT

Listing events without context makes the timeline hard to understand.



NOT DOCUMENTING

Poor notes and missing sources make your work hard to trust.



JUMPING TO CONCLUSIONS

Assumptions without evidence can lead to the wrong story.



NOT UPDATING

A timeline is never "done" until the incident is fully resolved.

SOC ANALYST TIMELINE TOOLKIT



SIEM

(Microsoft Sentinel, Splunk, QRadar)



ENDPOINT LOGS

(Windows, Linux, EDR)



NETWORK LOGS

(Firewall, Proxy, DNS)



CLOUD LOGS

(AWS CloudTrail, Azure Activity)



EMAIL SECURITY

(Microsoft 365, Proofpoint)



THREAT INTEL

(MITRE ATT&CK, VirusTotal)



FIND

the truth.



UNDERSTAND

the attack.



PROTECT

the organization.



IMPROVE

for the future.

A GREAT TIMELINE LEADS TO
A STRONGER DEFENSE!



PRACTICE. BE CURIOUS. STAY SHARP.
THAT'S WHAT GREAT SOC ANALYSTS DO!





READY TO LEVEL UP YOUR SOC SKILLS?



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guides, playbooks,
and cheat sheets
built by SOC
pros.



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challenges, and
scenarios to build
your skills.



SKILL TRACKING

Track your
progress and
earn badges
as you grow.



AMAZING COMMUNITY

Connect with
blue teamers,
ask questions,
and share wins.



TOOLS & RESOURCES

Curated tools,
templates, and
resources to work
smarter.



EVERYTHING YOU NEED TO GO FROM
LEARNING TO LEADING IN THE SOC.



PERFECT FOR:

✓ SOC Analysts (L1, L2, L3)

✓ Cybersecurity Students

✓ Incident Responders

✓ Aspiring Blue Teamers

✓ Threat Hunters

✓ Anyone Serious About
Blue Team Success

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